Unit 3 Stoichiometry Test - Teacher Version / Answer Key (verE)

Subject: chemistry
Course Code: SCH3U
Unit: Unit 3 Stoichiometry Test

Answer Key

1. [stoichiometry | written_response | 2 marks] 0.0898 mol Ca(NO3)2 approximately
2. [stoichiometry | written_response | 2 marks] 255 g K3PO4 approximately
3. [stoichiometry | written_response | 3 marks] 2.71 x 10^24 oxygen atoms approximately
4. [stoichiometry | written_response | 8 marks] Empirical formula C3H6O2; molecular formula C3H6O2
5. [stoichiometry | written_response | 4 marks] 14.2 g P4O10 approximately after using P4 + 5O2 -> P4O10
6. [stoichiometry | written_response | 10 marks] H2SO4 + 2NaOH -> Na2SO4 + 2H2O; 7.10 g Na2SO4 approximately
7. [stoichiometry | application | 6 marks] Pb(NO3)2 + 2KI -> PbI2 + 2KNO3; actual precipitate mass = 1.42 g; theoretical yield is about 1.89 g PbI2; percentage yield is about 75.1%

Student Paper Reference

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Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Foundational Stoichiometry Calculations

Section Plan: 3 questions | 7 marks

Instructions: Show all work. Include units and use appropriate significant digits.

1. Calculate the amount of substance in moles for 5.41 x 10^22 formula units of calcium nitrate, Ca(NO3)2(s). (2 marks)

2. Calculate the mass of 1.20 mol of potassium phosphate, K3PO4(s). (2 marks)

3. How many oxygen atoms are present in 2.25 mol of sulfur dioxide, SO2(g)? (3 marks)

Composition and Reaction Stoichiometry

Section Plan: 3 questions | 22 marks

Instructions: Show a complete stoichiometric setup. Write and balance any required chemical equation before solving.

4. A compound is 48.6% carbon, 8.16% hydrogen, and 43.2% oxygen by mass. Its molar mass is 74.1 g/mol. Determine the molecular formula of the compound. (8 marks)

5. In excess oxygen, phosphorus reacts to form tetraphosphorus decoxide, P4O10(s). Calculate the mass of P4O10 produced when 6.20 g of P4(s) is used. (4 marks)

6. Sulfuric acid reacts with sodium hydroxide to produce sodium sulfate and water. If 4.90 g of H2SO4(aq) reacts completely with excess NaOH(aq), what mass of Na2SO4(s) is produced? Write a balanced chemical equation for the reaction. (10 marks)

Lab Yield Analysis

Section Plan: 1 question | 6 marks

Instructions: Use the data provided to determine theoretical yield and percent yield. Show your setup clearly.

7. Use the lab data below to determine the percentage yield of lead(II) iodide, PbI2(s), produced when excess potassium iodide is added to lead(II) nitrate solution. Lab data: mass of Pb(NO3)2 used = 1.36 g; mass of clean, dry filter paper = 1.08 g; mass of dry precipitate and filter paper = 2.50 g. Write the balanced chemical equation for the reaction. (6 marks)